FIFA Player Data Viszalization Project Report

1. Introduction

The FIFA Player Dataset contains detailed information about professional football players, including their age, nationality, overall rating, market value, and skill attributes. This project focuses on analyzing and visualizing this dataset to extract meaningful insights about player performance and trends.

Data visualization helps in understanding large datasets easily by presenting them in graphical formats such as charts and plots.

2. Objective of the Project

The main objectives of this project are:

- ◦ To analyze player performance based on different attributes
- ◦ To identify trends in age, rating, and market value
- ◦ To compare players across positions and countries
- ◦ To present insights using visualizations

3. Problem Statement

This project aims to answer the following questions:

◦ Which countries produce the most football players?
◦ What is the ideal age range for peak performance?
◦ Which positions have the highest ratings? ◦
Who are the most valuable players?
◦ How are skills distributed among players?

4. Dataset Description

The dataset includes the following important features:
• Name - Player name
• Age - Player age
• Nationality - Country of player
• Overall - Overall rating
• Value - Market value
• Position - Playing position
• Skill attributes - Passing, shooting, defending, etc.

5. Data Cleaning and Preprocessing

Before visualization, the dataset was cleaned to ensure accuracy:

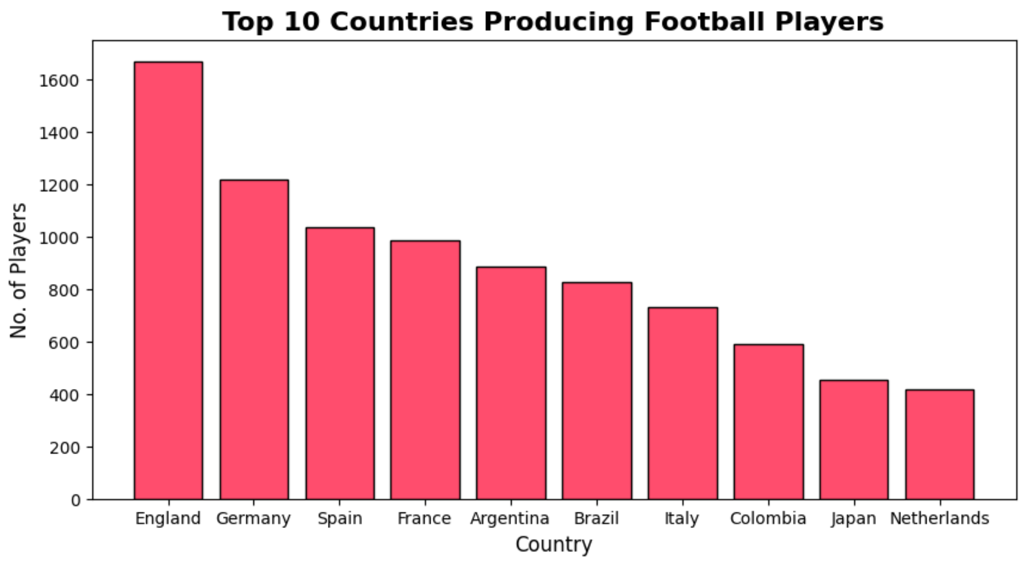
• Removed unnecessary columns (e.g., loaned_from)
• Handled missing values:
• Numerical → replaced with mean
• Categorical → replaced with mode
• Converted data types where necessary
• Ensured no null values remained

Importance

Data cleaning improves the reliability of analysis and prevents errors.

6. Data Visualization & Analysis

1. Top 10 Country Produce Football Player



The First chart shows that the comparison of no. of players of country and European countries dominate football production , also help club to choose players form this country

The dominance of these countries indicates strong football infrastructure, better training facilities, and well-established leagues. This uneven distribution also highlights that global football talent is concentrated in specific regions.

- The maximum no of players is from England
- Spain to Colombia their is decreasing order of small intervals

Why we use it:

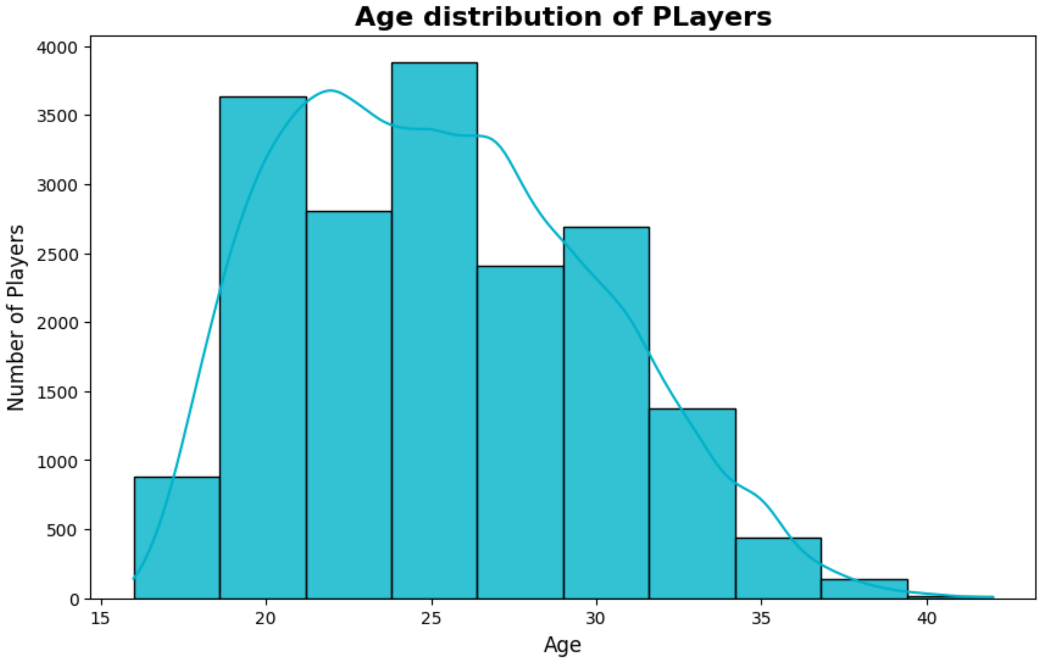
- To compare values between different categories
- Easy to see which category is highest or lowest

KEY REASON: BEST FOR COMPARISON

Insight

- Developed football nations produce more professional players
- This reflects the importance of infrastructure, coaching and exposure

2. Age distribution



The Second chart Shows that how players are distributed across age groups. And also help in find that a particular age people show more potential in game

This indicates that players are most active and competitive during this age range, Younger players are fewer due to limited experience, while older players decline due to physical limitations.

- Most players are between 20-30 years.
- Shows how many players fall into each age group
- Peak around 25 years

Why we use it:

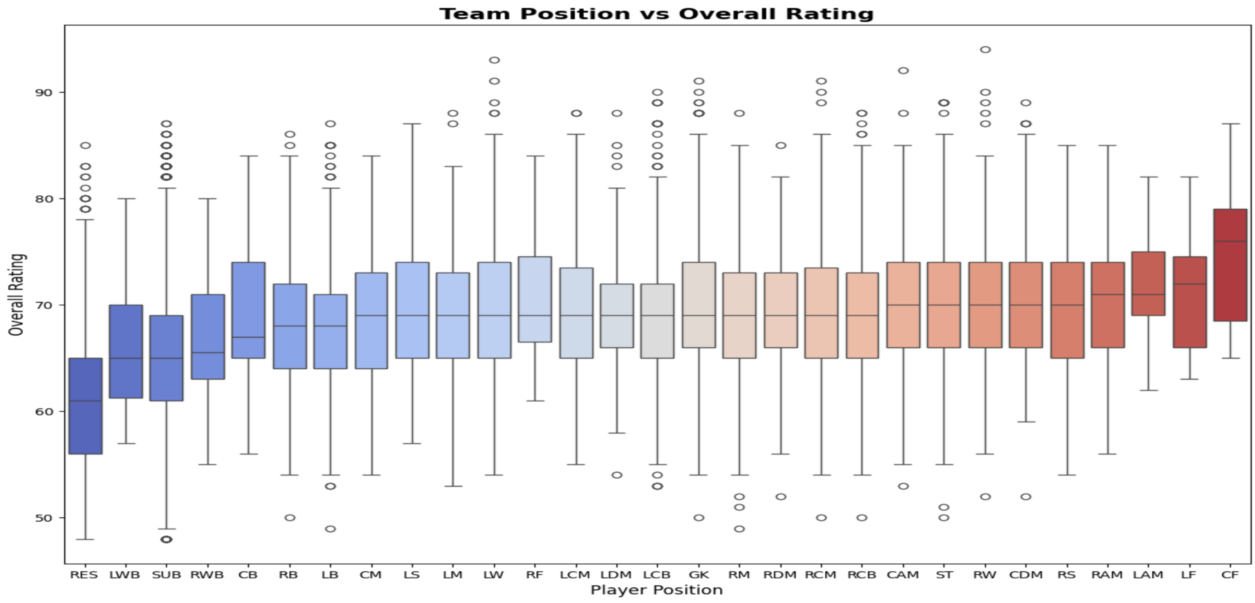
- To understand how data is distributed
- Shows frequency of values in ranges

KEY REASON: BEST FOR DATA DISTRIBUTION

Insight

- **players reach peak performance in mid -20s**

3. Team position vs Overall



The third chart shows that the player's rating according to their position , also help in deciding the position for the player in other words compare rating across position

It shows that attacking players generally have higher ratings, while defensive players have more consistent performance. The presence of outliers indicates exceptional players in certain positions.

- RES players often have lower ratings
- Centre Forward (CF) players often have higher ratings.

Why we use it:

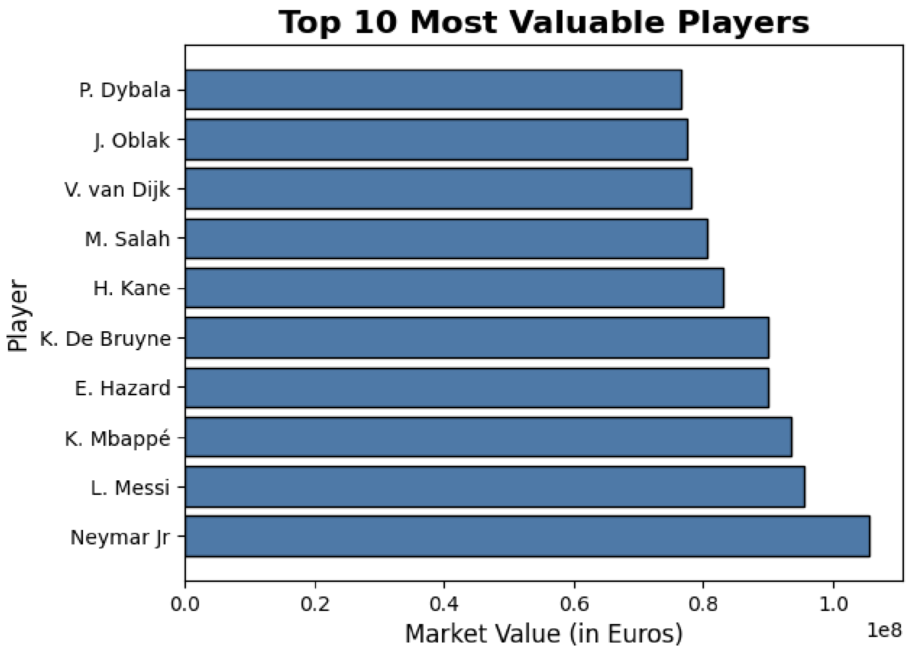
- To show spread of data
- Helps detect:
  - Median
  - Outliers
  - Variation

KEY REASON: BEST FOR VARIATION & OUTLIERS

Insight

- **different positions have different performance demands**

4. Top 10 Valuable Players



The fourth chart shows that the top 10 valuable players and shows high rated players have higher market value the other players

It shows that a small group of elite players holds significantly higher value than others. This is mainly due to their high performance, popularity, and future potential.

The addition of value labels makes the chart more informative, showing that a small group of players dominates the market value distribution.

- P. Dybala has lowest market value
- Neymar jr. has highest market value

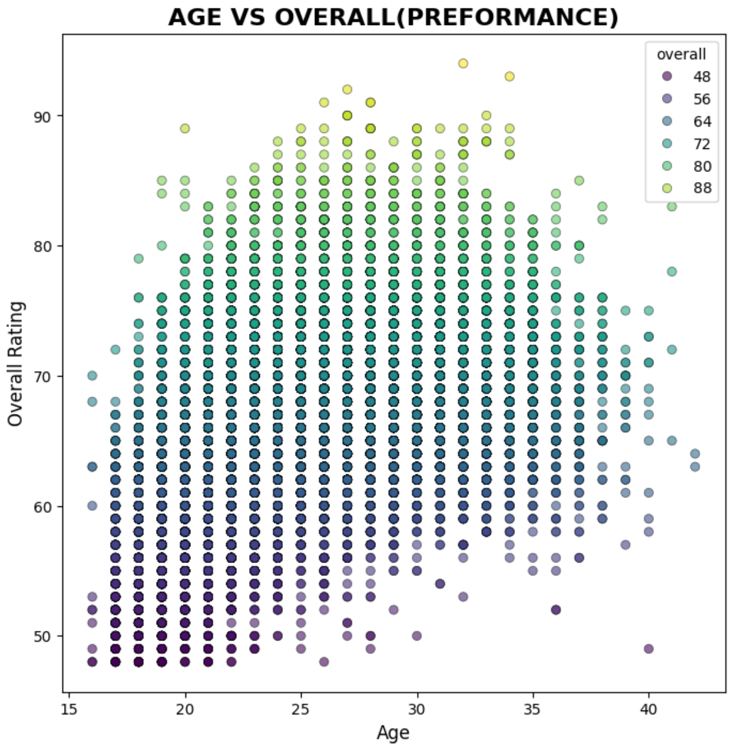
Why we use it:

- To compare values between different categories
- Easy to see which category is highest or lowest
- Horizontal bar chart of highest-value players.

Insight

- Footballs follows a ‘star player economy’

5. Age vs Overall



The fifth chart shows that the relation between age and performance, they show how age affects performance it’s means that any particular age group have high potential then others Age group

It also indicates that player performance improves with age up to a certain point (around 25-30 years, after which it stabilizes or declines. This reflects the typical career lifecycle of football players.

- The 20-25 age players has highest rating
- While 35 - 40 age players has lowest rating

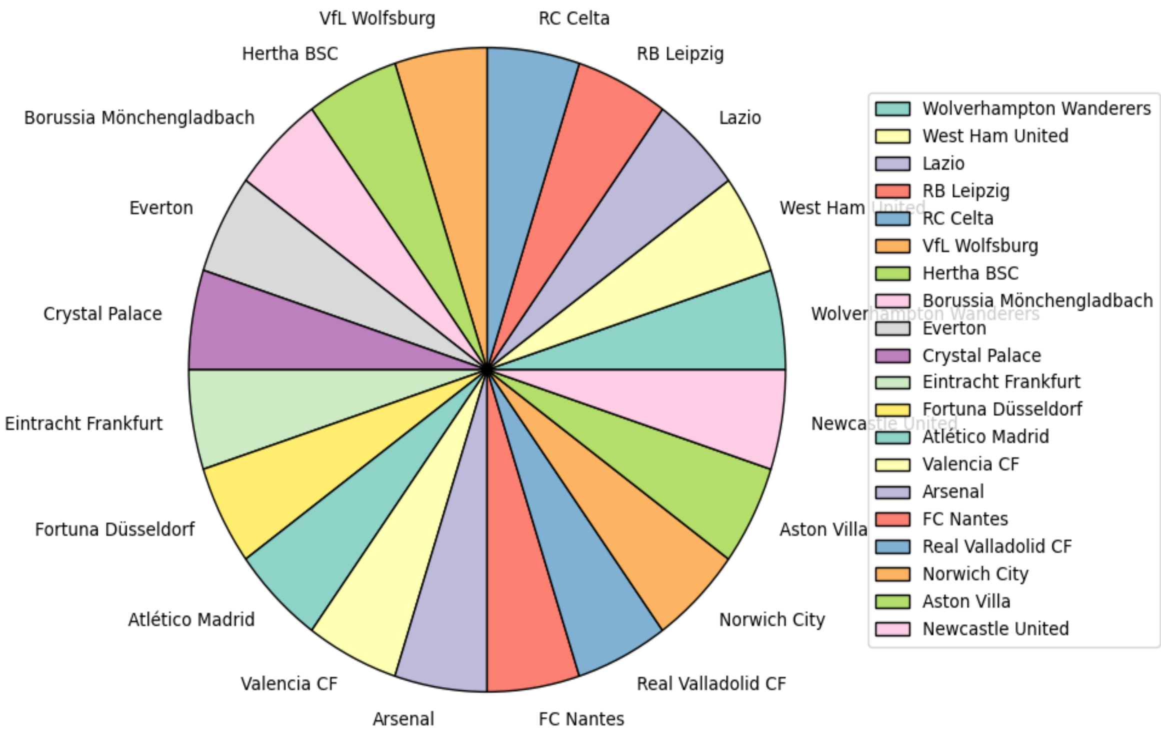
Why we use it:

- To find relationship between two variables
- Shows how age affects performance

Insight

- career follows a growth, peak, decline pattern

6. TOP 20 clubs give more players



The Sixth chart shows that the distribution of clubs according to larger share of players, and also, this visualization helps in understanding the structure of player distribution across clubs and provides valuable insights for further exploratory data analysis.

The chart shows that 20 club has equal share of players

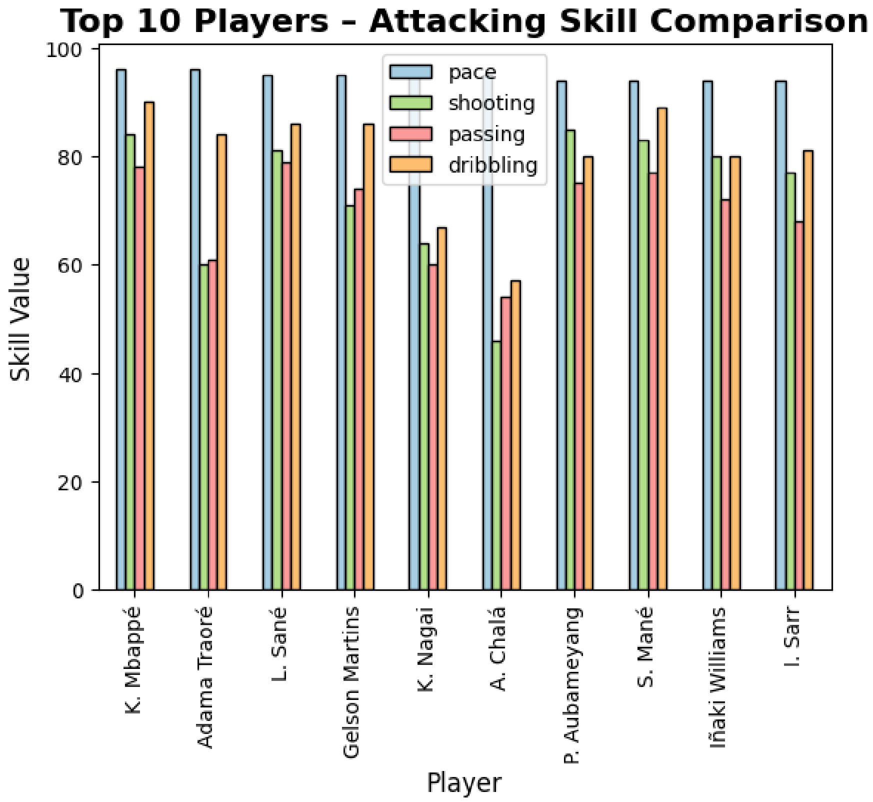
Why we use it:

- To show club distribution
- To show percentage of sharing of players

Insight

- Strong club systems lead to more player development
- Attacking players dominate this category
- Skills are relatively balanced across top players

7. Top 10 Players- Attacking Skill Comparison



This chart compares key attacking attributes (pace, shooting, passing, and dilbling) of the top 10 players. By selecting players based on total attacking score.
the visualization highlights the most offensively skilled players. It shows that elite players tend to perform consistently across multiple attacking attributes rather than excelling in just one.

- Top players perform well across multiple skills
- Balanced players are more valuable than specialists
- Skills are interdependent

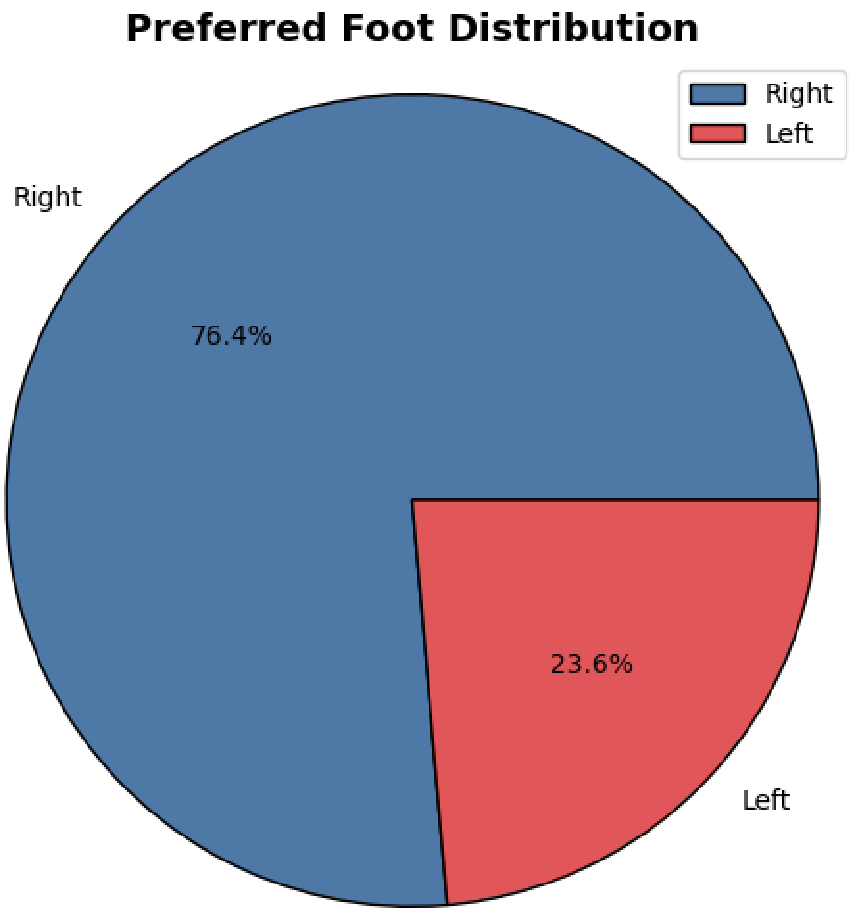
Why we use it:-

- Compare multiple skills together
- Helps understand overall attacking ability
- Helps find specialists vs balanced players

Insight

- ELITE ATTACKERS ARE MULTI-SKILLED.

8. Preferred Foot Distribution



This chart shows the number of players based on their preferred foot.

- Majority of players are right-footed
- Left-footed players are significantly fewer
- This creates a natural imbalance in gameplay

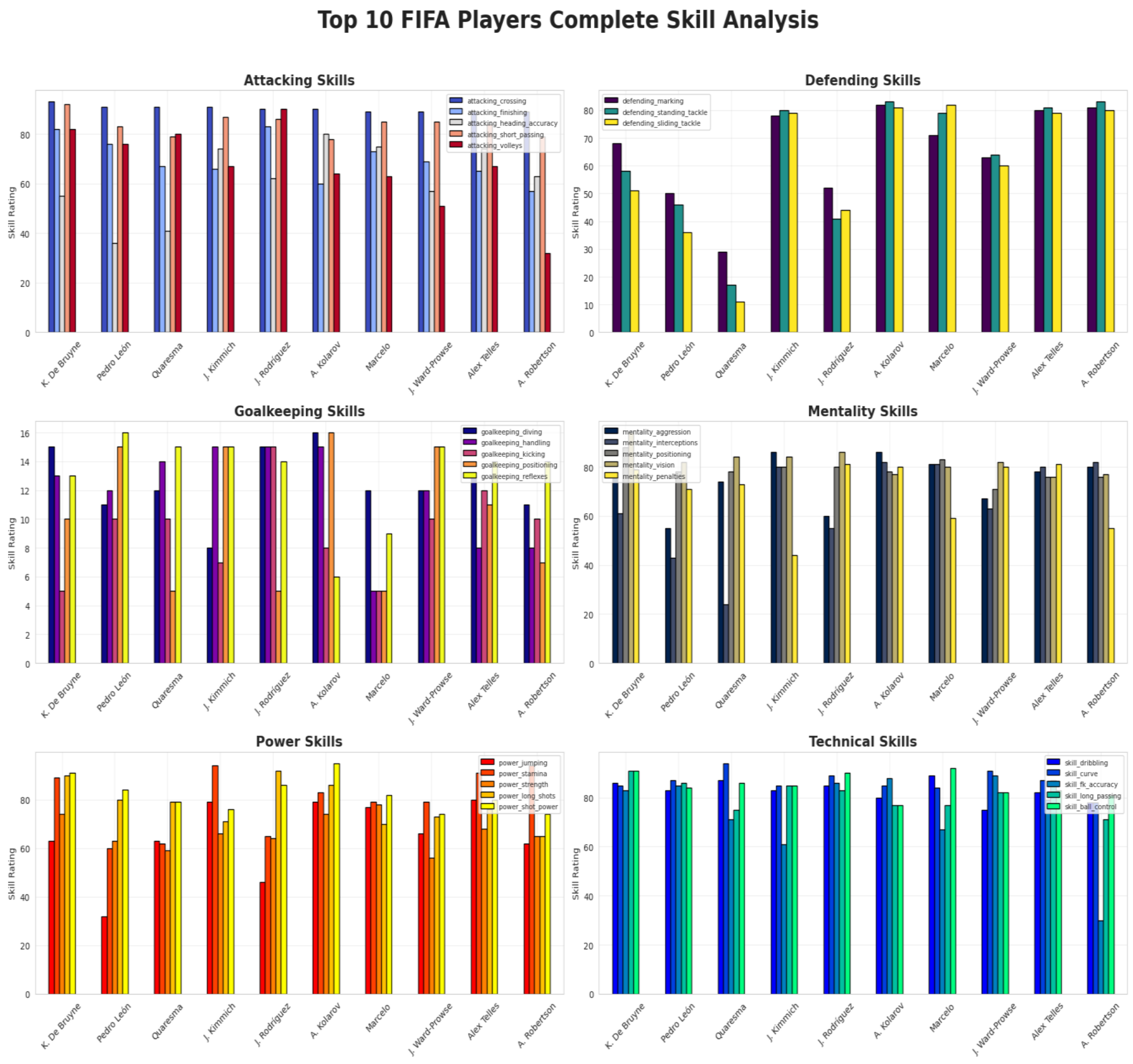
Why we use it :-

- Shows whether players prefer **left or right foot**
- Helps understand player behavior and style

Insight

- left footed players are rare and valuable strategically

9. Top 10 FIFA players skill comparison



This chart shows that elite players are strong across multiple skill categories . attacking and technical skills such as passing, dribbling, and finishing are the most dominant among top players

Defensive and goalkeeping skills are more position specific and vary defending on the player's role. Power and mentality attributes support overall performance defending on the player's role. Power and Mentality attributes support overall performance by improving physical strength and decision – making

- **Attacking and technical skills** (passing, dribbling, finishing) are the strongest
- **Defending and goalkeeping skills** are lower for non-specialized players
- **Mentality skills** (vision, positioning) are consistently strong
- **Power attributes** (stamina, strength) support overall performance

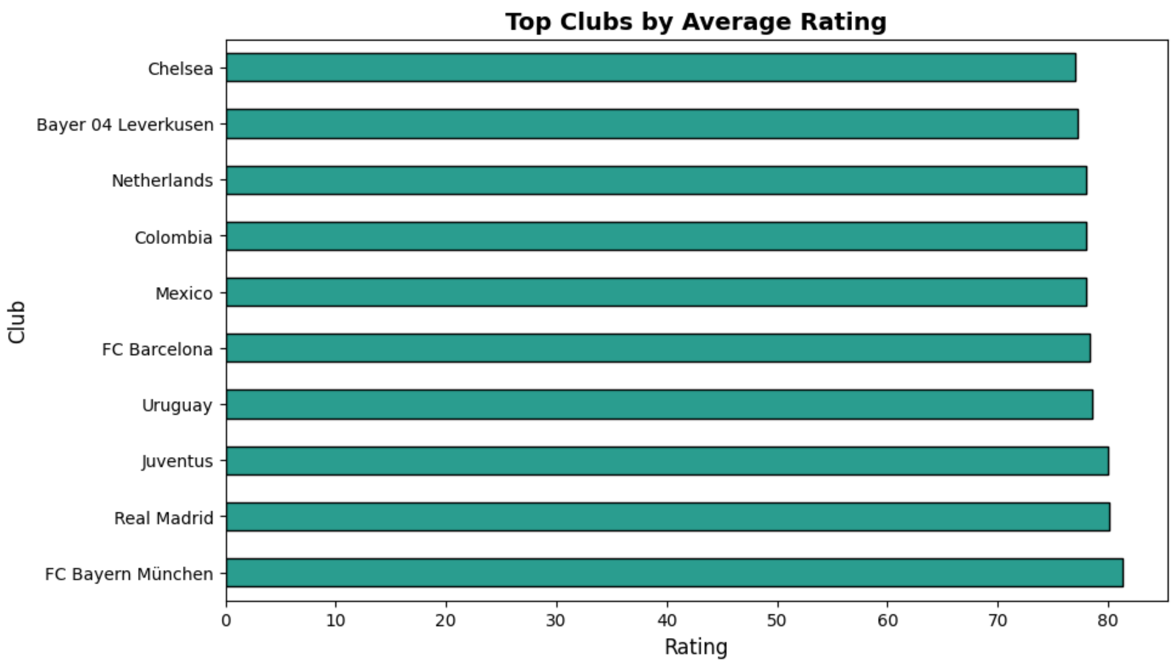
Why We Use it:-

- To compare multiple skills at once
- To identify strengths and weaknesses of players
- To understand overall player performance

Insight

- Elite players are **multi-dimensional and well-balanced**
- Performance depends on a combination of:
  - Technical ability
  - Physical strength
  - Mental intelligence
- Specialization varies based on player position

10. Top Clubs by Average Rating



This chart displays the top 10 football clubs based on the **average overall rating** of their players. Instead of counting the number of players, this analysis focuses on the **quality and performance level** of players within each club.

- Some clubs maintain consistently high-rated squads
- Clubs with higher average ratings usually:
  - Have elite players
  - Strong financial support
  - Better training infrastructure

- The difference in ratings between clubs is relatively small, showing strong competition among top clubs

Why We Use it:-

- To compare club performance based on player quality
- To identify clubs with the strongest squads
- To analyze overall team strength using average ratings

Insight

- Average rating reflects the **overall strength of a club**
- Clubs with highly rated players are more likely to perform better in competitions
- This analysis highlights the importance of squad quality over quantity

7. Conclusion

This project successfully analyzed the FIFA player dataset using various data visualization and analytical techniques. Through charts, graphs, and comparisons, important patterns related to player performance, age, market value, clubs, and skills were identified.

The analysis showed that most players are between **20–30 years old**, which represents the peak performance phase in football. Market value was found to be strongly influenced by overall rating, age, and player potential. A small group of elite players dominated both ratings and market value, reflecting the competitive nature of professional football.

Country-wise and club-wise analysis revealed that football talent is concentrated in specific countries and top clubs due to strong infrastructure, training systems, and player development programs. Skill comparison charts demonstrated that elite players are multi-skilled and perform strongly across attacking, technical, physical, and mental attributes.

The project also highlighted that different positions require specialized abilities:

- Attackers excel in pace, dribbling, and shooting
- Defenders perform better in tackling and marking
- Goalkeepers have unique specialized skills

Overall, this project demonstrates how data visualization can simplify complex sports data and help uncover meaningful insights for football analysts, scouts, coaches, and decision-makers.

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